

CASE STUDY PROJECT DESCRIPTIONS

All students must choose to work on a case study project in groups of two students. The requirement to work in groups of two students is rigid and is based on previous experiences with teaching modeling, implementation, and analysis using software, and it will increase learning outcomes.

Two case studies are offered, and you must select one case study to work on. When the group has chosen the case study on which they would like to work, the selected choice must be recorded by the teaching team to cater to the best possible planning of assistance for the groups (and to ensure that everyone is assigned to a project 😊).

You and your project partner sign up in DTU Learn. The registration will be open from Tuesday June 16th.

The two case study projects are described in the following pages and fall within the following topics:

- 1) Healthcare
- 2) Manufacturing

Each group must plan and carry out their work for all case studies according to the “*General simulation methodology*” (cf. Rossetti, 2016), covering at least the mandatory requirements mentioned in the case studies.

Case study project hand-in:

Please refer to the last part of this document for precise instructions on how to hand in the project.

Deadline:

The project deliverables must be handed in via DTU Learn no later than **Friday, June 26th at 18:00**.

1. Healthcare: Apheresis for CAR-T cell production

Apheresis is a process, in which the patient's blood is collected in a controlled manner to extract specific components, such as immune cells (Qayed et al., 2022). Therefore, apheresis serves as an entry point for cell and gene therapies, such as CAR-T cell treatments for cancer patients (Buechner et al., 2018). Apheresis is usually conducted in a hospital setting before the collected apheresis product is sent to a pharma manufacturer for production of CAR-T cells.

CAR-T therapy is a cancer therapy that shows high efficacy for treating haematological cancers (7% of all cancers) and is a one-off treatment, making it patient friendly. However, only ~18% of eligible patients receive CAR-T therapy (IQVIA, 2025) due to operational barriers: high costs (€300,000–\$475,000 per infusion) (Heine et al., 2021), specialized and high resource requirements, process complexity from product-specific variations, manufacturing time of 3-4 weeks, and coordination demands between apheresis centres, hospitals, and manufacturers. This results in about 1 in 4 patients dying while waiting for treatment (Al Hadidi et al., 2023), so making treatment process – including the apheresis process – as efficient and quick as possible is of high importance. Additionally, apheresis should be more efficient because indications for CAR-T cell therapy are expanding towards solid tumours and autoimmune diseases, so demand for apheresis is expected to increase.

In this study project, the management of a big hospital in Germany has asked you to look at their apheresis set-up and study the operational performance using discrete event simulation. Management is particularly interested in identifying strategies to meet the future increased demand for apheresis whilst running the hospital with the highest patient care and good working conditions.

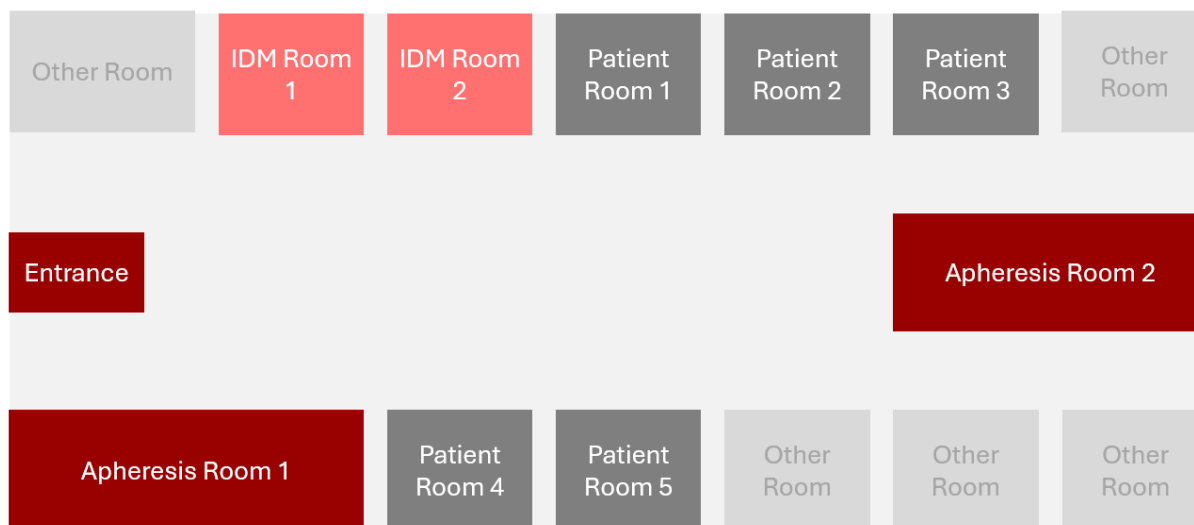


Figure: Layout of hospital with IDM and apheresis rooms. Each apheresis room contains one apheresis machine. In each patient's room, there are two beds.

The process can be described as follows. First, the patient is hospitalized (**seizes a bed**) and must get checked for infectious markers a day prior to apheresis, so-called **IDM check-up**. If no bed is available, the patient should come back the next day. The IDM check-up occurs during

normal **working hours** (8am – 4pm, Mo-Fri). This IDM check-up is performed at the hospital by a **nurse** in one of the **two available IDM rooms** (one patient allowed per IDM room). About **20% of patients will have a positive IDM test result** (they have an infection and cannot do the apheresis in that state) and will receive therapy (resources for therapy will not be modelled in this simulation, no requirement for a bed) for two weeks. After the therapy, these patients repeat the IDM test (and then succeed in receiving a negative IDM result, i.e., no infection).

For patients with a negative IDM result, they can proceed to the **apheresis check-up** the day after the IDM check-up. The apheresis check-up, apheresis, and monitoring require specifically trained staff called “DRK” (DRK: German Red Cross) who come to the hospital to perform the apheresis process steps. *Hint: DRK staff leave at 2:30pm, so all processes requiring DRK staff should be done by then. The steps of apheresis check-up, apheresis, and monitoring all occur in the apheresis room. The apheresis room is large and can host multiple patients at the same time.*

The **apheresis check-up** precedes **apheresis** and is performed by **one DRK doctor** in the apheresis room (patient in bed in patient room is moved to the apheresis room – this process does not have to be modelled).

After this, **apheresis** is performed by 1 DRK doctor and 1 DRK nurse. One apheresis machine is used per patient for the entire duration of the apheresis. Note that the apheresis machines fail on average every 48 hours and will have to be repaired by a DRK operator. The time for the repair can be assumed by a lognormal distribution with a minimum repair time of 20 min, mean repair time of 60 min and standard deviation of 15 min. *Hint: AnyLogic has a different format for lognormal distribution.*

Lastly, the patient is **monitored** directly after apheresis **by one DRK doctor and DRK nurse**. It is to be noted that the DRK doctor can **supervise up to 10 patients** simultaneously across apheresis and monitoring. During monitoring, most common symptoms that patients may experience are fatigue, dizziness, and low blood pressure. The majority of patients meet the conditions for discharge (stable vital signs, feel well enough to go home) after monitoring ends and can go home. For this case study, we assume that all patients can go home after monitoring and the beds are given to the next patient. You can assume that the simulation model finishes at this step. In practice, after monitoring, a courier from the pharmaceutical company producing the CAR-T cells is sent to pick up the apheresis product to transport it to the pharmaceutical CAR-T production site. *Hint: Think about real treatment circumstances, e.g., patients can develop serious side effects after a blood draw.*

Currently, between **one and three patients** enter the apheresis process **every three days**.

Table 1: Number of available resources (rooms, beds, machines)

Number of IDM rooms	2
Number of beds	10
Number of apheresis machines	2

Table 2: Shifts and number of assigned employees

Resource	Days available	Start of shift	End of shift	Amount
Nurse	Every day	6 am	10 pm	10
DRK doctor	Mo and Thu	6:30 am	2:30 pm	1
DRK nurse	Mo and Thu	6:30 am	2:30 pm	3
DRK operator	Mo and Thu	6:30 am	2:30 pm	1

Mandatory requirements

Based on the description given above you should carry out a simulation study of the operational performance of the apheresis. This includes the following tasks.

- (1) Formulate the problem (e.g., flowcharting), define suitable performance metrics (KPIs), and build a conceptual model of the apheresis process incl. IDM check-up.
- (2) Build and verify a simulation model in AnyLogic. *Hint: Choose appropriate time units.*
- (3) Define appropriate input data (use the data provided in the Excel sheet) and validate the simulation model.
- (4) Design an experimental study for analyzing the apheresis process. You may consider assigning priority to patients that undergo the IDM checkup for a second time (if their first IDM result was positive).
- (5) Draw managerial insights to present to management.

Specific issues to be examined and/or discussed are the following:

- (6) **Bottlenecks:** Identify bottlenecks in the current setup and make suggestions for improvement. Use your simulation model to underpin your suggestions.
- (7) **Sensitivity analysis:** Perform a sensitivity analysis investigating different arrival patterns of the patients (i.e., rising patient demand). What is the maximum number of patients that can be processed in the current set-up? *Hint: You might use a multi-factorial test design (i.e., changing several parameters at the same time).*

Further analyses MIGHT focus on the following:

- (1) **Increased demand:** Your manager wants to be able to perform at least 40 apheresis per month to meet the expected increased demand arising from new indications being approved for CAR-T cell therapy. Suggest possible changes to the setup of the apheresis (e.g., number of available apheresis machines, additional shifts by DRK staff, additional DRK staff members) and assess the implications of these changes by your (extended) simulation model. Discuss the value of the proposed changes from a management point of view. *Hint: Consider normal hospital staff becoming certified for apheresis processes.*

- (2) **Ambulatory scenario:** Currently, patients seize a bed before IDM and continuously hold it until the end of monitoring. The manager has observed that patients are well enough to go home between treatment steps, meaning beds are occupied unnecessarily during waiting and queuing periods. The manager proposes that patients go home between treatment steps, specifically after the IDM checkup. Patients are then readmitted to a bed for the apheresis checkup. Assess how different prioritization rules for the beds affect system performance, e.g., FCFS, health-based prioritization, prioritization based on IDM result, progress-based prioritization (patients further along in the process receive higher priority for the next available bed).
- (3) **Clinical constraints:** A clinical constraint is that IDM results are only valid for 5 days – patients waiting longer must repeat the IDM again (patient joins the main flow of IDM, apheresis check-up) before the patient repeats apheresis. *Hint: You may want to explore the exit on timeout function.*
- (4) **Industry 4.0:** Investigate Industry 4.0 (Smart Manufacturing) for the apheresis machines, specifically 'Predictive Maintenance,' and suggest possible solutions to reduce malfunctions. What is the potential outcome of implementing preventive maintenance?

References

- Al Hadidi, S., Szabo, A., Esselmann, J., Hammons, L., Hussain, M., Ogunesan, Y., Thalambedu, N., Khan, F., Sethi, J., Janardan, A., Radhakrishnan, S. V., Thanendrarajan, S., Schinke, C., Dhakal, B., Janz, S., Chhabra, S., D'Souza, A., Zangari, M., van Rhee, F., & Mohan, M. (2023). Clinical outcome of patients with relapsed refractory multiple myeloma listed for BCMA directed commercial CAR-T therapy. *Bone Marrow Transplantation*, 58(4), 443–445. <https://doi.org/10.1038/s41409-022-01905-1>
- Buechner, J., Kersten, M. J., Fuchs, M., Salmon, F., & Jäger, U. (2018). Chimeric Antigen Receptor-T Cell Therapy. *HemaSphere*, 2(1), e18. <https://doi.org/10.1097/HS9.0000000000000018>
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- IQVIA. (2025). *Achieving CAR T-cell Therapy Health System Readiness*. <https://www.iqvia.com/insights/the-iqvia-institute/reports-and-publications/reports/achieving-car-t-cell-therapy-health-system-readiness>
- Qayed, M., McGuirk, J. P., Myers, G. D., Parameswaran, V., Waller, E. K., Holman, P., Rodrigues, M., Clough, L. F., & Willert, J. (2022). Leukapheresis guidance and best practices for optimal chimeric antigen receptor T-cell manufacturing. *Cytotherapy*, 24(9), 869–878. <https://doi.org/10.1016/j.jcyt.2022.05.003>

Formalities

The results of your work must be communicated via a written report (allowing appendices to be used for inclusion of relevant figures and tables) and via a video presenting the simulation project. **PLEASE REFER TO THE INSTRUCTION AT THE END OF THIS DOCUMENT.**

2. Manufacturing

FermaCore Labs is a fast-growing Scandinavian manufacturer of functional tablet products. Demand has increased steadily and management has launched a 2030 plan to maximize output within the existing production capacity for as long as possible before committing to an expansion (new equipment and packaging lines have long lead times).



The production system combines batch-type operations and rate-based operations, making it well suited for discrete-event simulation. Management has requested an analysis to assess current performance, identify bottlenecks and evaluate improvement initiatives.

FermaCore produces three final tablet products: **CoreTab** (standard tablet), **PlusTab** (film-coated tablet) and **MaxTab** (film-coated tablet). To keep the model tractable, all products follow the same routing structure through the line; differences between products are captured only in the coating and a packaging changeover time when switching products on a packaging line. All other processing times and yield assumptions are identical across products in the baseline model.

FermaCore operates with a weekly shipping cycle, meaning that all finished goods produced within a given week are consolidated into a single weekly shipment to customers. Management has provided a draft production plan for the simulation period, specifying which product variants are to be produced and when. The batch size is fixed and defines the ingoing active ingredients weight. The overall volume commitments in the plan must be respected. Due to operational constraints, an order may not deviate more than one day from its originally planned production date.

Description of the setup

The manufacturing line receives product in process-batches. In the baseline model, one batch corresponds to 150 kg of a pre-weighed blend of active ingredients (powder form) and 75 kg of excipients ready for processing. Each produced tablet contains 500 mg active ingredients with a total weight of 750 mg including the excipients. The batch sizes are, for the base case, assumed to be fixed throughout the entire production process and cannot be altered. Due to strict pharmaceutical traceability regulations, individual batches must remain fully separated at all times - mixing of batches at any stage of production is not permitted.

Product variety is created by converting this base material into three final tablet variants in the coating step. It is important to note that prior to coating, all batches are physically identical; a batch only becomes a specific product variant (CoreTab, PlusTab, or MaxTab) from the coating step onwards, including all subsequent downstream operations. Each batch enters the line as a single entity and the selected variant depends on the production plan. The line consists of six main operations, these are described below.

First, the batch undergoes *Premix/Blending*, where active ingredients and excipients are mixed to form a homogeneous powder blend suitable for downstream processing. This operation is performed on two blending units operating in parallel: an *old* unit (slow) and a *new* unit (fast).

Since blending time depends on the formulation and equipment performance, the processing time in the *new* unit is modeled as a triangular distribution with parameters $\min=3.5$, $\max=4.5$ and $\text{mean}=4.0$ hours per batch. For the *old* unit, the processing time is modeled as a triangular distribution with parameters $\min=4.5$, $\max=6.5$ and $\text{mean}=5.5$ hours per batch. Batches are routed to either unit based on availability.

In addition, the premix/blending units can be subject to failures. The failure frequency is specified in the *FermaCore_input_data.xlsx*, while the repair duration should be found using the distribution fitter. If a failure occurs while a batch is being processed in premix/blending, the entire batch is assumed to be destroyed and therefore scrapped. This failure model applies to both the *old* premix/blending unit and the *new*, though the frequency differs.

Second, the blended material is processed in a *Granulation* unit, which converts the blend into uniform granules suitable for tablet pressing. The processing time is modeled as a normal distribution with mean 1.0 hour and standard deviation 0.10 hour. In total, one granulation unit is available.

Third, the granules undergo a *Drying/Conditioning* step, where moisture is reduced to achieve stable flow and compression behavior. The processing time is estimated from the time observations provided in the file. In total, one drying/conditioning unit is available. After a batch leaves the unit, a fixed 2.0 hour cleaning process is required before the unit can receive the next batch. The yield for this step is assumed to be 100% in the baseline model.

Fourth, the batch undergoes *Tablet Compression*, where the dried granules are compressed into solid tablets using high-pressure tablet presses. This step transforms the granulated material into uniform tablets with defined weight, shape and mechanical strength. It takes the machine 2 hours to process one batch of 225 kg, reflecting the highly automated and rate-controlled nature of the compression process. The batch occupies one of three compression units.

Fifth, tablets may be processed in a *Coating* unit to create product variety. The CoreTab product skips this step (uncoated, no dosing). For PlusTab and MaxTax a firm-coat is applied. The processing time can be modeled using a triangular distribution with $\min=1.5$, $\max=2.5$ and $\text{mean}=2.0$. In total, one coating unit is available. After a batch leaves the coating unit, a fixed 1.0 hour cleaning process is required before the unit can accept the next batch.

Finally, finished product is sent to *Packaging*, performed on three parallel lines. Processing is rate-based: assume each line processes approximately 30,000 tablets per hour. The tablets are packed into bottles of different sizes, which can be seen in *FermaCore_input_data.xlsx*. If there are not enough tablets remaining in a batch to fill a complete bottle, those leftover tablets are scrapped. When a line starts a new batch, a fixed setup time of 0.50 hours is required. Furthermore, when a line switches from one product type to another, an additional fixed 0.50 hour changeover time is incurred. Both the setup and changeover is performed by the cleaning team.

Optionally, the tablet packaging lines can be subject to downtime. The downtime frequency is specified in *FermaCore_input_data.xlsx*, while the repair duration should be found using the distribution fitter. The three tablet packaging lines are assumed to have identical downtime behavior in the baseline model. Before packaging begins, a visual inspection for

quality control is performed and is therefore included as part of this operation. This inspection is accounted for by assuming a small yield loss due to defective or non-conforming tablets identified prior to packaging.

All cleaning operations, changeovers and maintenance activities across the system are assumed to be performed by a single shared cleaning team. This implies that cleaning-related activities compete for the same limited resource, potentially introducing additional waiting times between production batches.

Mandatory Requirements

Based on the description given above, you should carry out a simulation study of the operational performance of the manufacturing setup. This includes:

- (1) Problem formulation, including the establishment of performance metrics and building a conceptual model of the production.
- (2) Simulation model building, including model translation, input data modeling and verification and validation.
- (3) Setting up the experimental design and carrying out the simulation analysis, including performing preliminary test-runs, final experiments and the final analysis of the results.

Specific Issues to be Examined and Discussed

- (1) The FermaCore management would like to know where the bottlenecks are in the present setup with the current data. Use your simulation model to answer this question. Choose and highlight relevant KPIs.
- (2) Based on your bottleneck analysis, suggest improvements to the production line which will optimize the performance of the line. You are free to suggest changes to process times, number of resources etc. Hint: consider the sequencing of product batches.
- (3) The management would like you to examine the operational performance of the present setup towards 2030 using the chosen KPIs. Where are the bottlenecks with these data? Which combination of improvements maximizes the performance? Again, consider the sequencing of the batches.

Non-mandatory requirements – Ideas for further analysis

Employees from the quality assurance department have observed a correlation between unusually high scrap and a sudden decrease in press force and/or filling depth during the tablet pressing operation. Employees estimate that if it is possible to detect the deviation before the press drifts out of specification, it would be possible to reduce the scrap rate from 10% to 4%. Therefore, FermaCore management is considering an offer from an equipment vendor to install sensors that monitor the tablet press in real time. This would mark the management's first significant step towards efficiency improvements through ideas from Industry 4.0.

- (1) Investigate Industry 4.0, specifically *predictive/preventive maintenance* and suggest possible solutions whereby the use of sensors can lead to reduced scrap. What is the potential outcome of implementing predictive/preventive maintenance?

Input data

One agent represents one process-batch. In the baseline model, one batch corresponds to 150 kg of the pre-weighed active ingredients (75 kg of excipients are added in the Premix/Blending step). Each batch is assigned a product variant (CoreTab, PlusTab, or MaxTab) according to the production plan.

More detail on data and times can be found in FermaCore_input_data.xlsx.

Formalities

The results of your work must be communicated via a written report (allowing appendices to be used for inclusion of relevant figures and tables) and via a video presenting the simulation project. **PLEASE REFER TO THE INSTRUCTION AT THE END OF THIS DOCUMENT.**

GUIDELINES FOR CASE STUDY PROJECT DELIVERABLES

GUIDELINES FOR CASE STUDY PROJECT DELIVERABLES

This guideline document lists the formal requirements for the deliverables for the case study projects.

Each group delivers:

1. A technical report including relevant figures, tables, spreadsheets, and illustrations (PDF format) (see next page for a suggested format of the technical report)
2. The AnyLogic project file, data files, etc. (zipped). Please do not include all binaries and data output files. Ensure you only upload a zipped file containing the AnyLogic model and your input data (please do not include the Log files, which can be very large).

Each student delivers:

1. An individual video that presents the simulation project and results obtained (standard video format, MP4, etc.).
 - Please note that you have to present your own slides/give your own presentation. Copying your presentation from you group partner will be considered as plagiarism!

Handing in the deliverables

An assignment will be available in DTU Learn, allowing each group to hand in both the group deliverables and the individual videos.

The deadline for handing in the deliverables is **Friday, June 26th, at 18:00.**

In the following, more information on the expectations of the deliverables is provided. If you have any questions, do not hesitate to ask! 😊

*Author 1: Name and student ID**Author 2: Name and student ID*

Technical report

In the technical report, you and your group partner must document your simulation project with figures, graphs, tables, etc. Following the phases and activities as part of the “General simulation methodology” by Rossetti (2016).

Please remember not to “overdo” it. Consider what to include and what not to include. The maximum page limit of the technical report is 15 pages (**this includes ALL pages – also the front page**)!

Suggestion for the structure of the report:

- 0) Front page (student names, student IDs and group number)
- 1) List of contents
- 2) Statement of collaboration
 - a. List which other groups you collaborated with (if applicable) or list that you did not collaborate
- 3) Performance management
 - a. List your KPIs – argue why these were chosen
- 4) Conceptual modeling
 - a. Flow Chart of the system under consideration
 - b. Activity Diagram of your conceptual model
- 5) Document assumptions
 - a. Briefly explain which specific assumptions you made (could also be part of the Activity Diagram mentioned above)
- 6) Input modeling
 - a. Document your input modeling
- 7) Verification and validation (calibration)
 - a. Verify your model implementation in AnyLogic
 - b. Validate and calibrate your AnyLogic model
- 8) Output analysis
 - a. Decide on the minimum required replications needed to make the conclusions trustworthy
- 9) Experimental study
 - a. Design your experimental study, i.e., explain why these scenarios were chosen (e.g., use a scenario tree to illustrate the relations between the various scenarios)

- b. Present and discuss the results of the scenarios using graphical illustrations/figures comparing KPIs across scenarios
- c. However, remember you do not have to include everything you have done

10) Conclusions and insights

Individual video presenting the simulation project

The final deliverable is a recorded video presenting your project. The **video is an individual hand-in and is, therefore, your personal communication** of your project. You can make use of the same results and figures/graphics, and, of course, the same AnyLogic project as your group partner, BUT it is important that you each do your own separate communication of the project. So you are not allowed to share the same PowerPoint/slides and use the exact same words for communicating the project.

The video must present what you have been doing during the project and the results of your simulation project.

Suggestion for a “story-line” for your video:

- 1) Introduce yourself - **cameras must be on – at least for the intro.**
- 2) Introduce the objective of the simulation study
- 3) Explain the model (point to the various parts of the model – either on the output window or in the model “main” window)
- 4) Demonstrate the model (either simple functionality or extended functionality – you decide)
- 5) Present your main findings and conclusions of the project

Please note that you have to present your own slides/give your own presentation. Copying your presentation from you group partner will be considered as plagiarism!

Technical notes:

- The video does not have to be very high-quality in terms of visualization of the simulation model. A 3D visualization can indeed, in some cases, be an appealing way to present your simulation model, but it is NOT required. A 2D visualization or explaining the model from the “main” window may also work.
- Write your name and student ID on the title page/intro/opening of the video. We recommend that you use Zoom or Teams to record. The video should be in a standard video format, preferably in MP4.
- The length of the video **cannot exceed 5 minutes**. Do not use “speed up”!